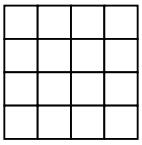
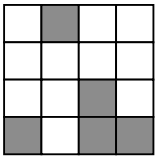
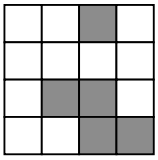
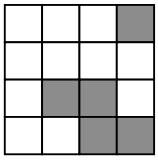
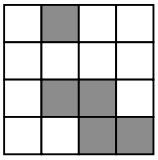
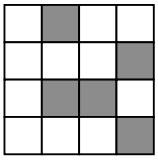
Math Kangaroo Practice

Grade 3-4 · 15 questions · harder · seed 15897

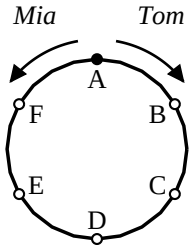
Part A: Each question is worth 3 points

1. Maria colours exactly 5 cells of a 4 × 4 grid grey. Five friends guess which cells Maria coloured: the patterns A to E. Maria says: "One of you is right. Each of the others guessed exactly 4 cells correctly." Which pattern did Maria colour?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

2. Tom and Mia draw a big circle in the sand and write the letters A, B, C, D, E and F on it. They start at the same time from point A. Tom runs clockwise and Mia runs anticlockwise around the circle (see picture). They meet for the first time at point B, then at C, at D, at E, at F and then again at point A. How many times has Mia run around the circle by then?



A. 4 times ***B.*** 5 times ***C.*** 6 times ***D.*** 7 times ***E.*** 12 times

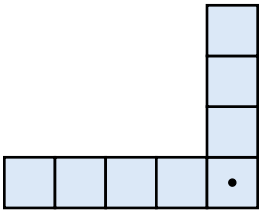
3. Four children, Mia, Lea, Anna and Emma, run a race. There are no ties. Mia is not the fourth. Emma is not the second. Emma is not the third. In which order can they be?

A. Mia, Emma, Lea, Anna ***B.*** Emma, Lea, Anna, Mia ***C.*** Emma, Anna, Lea, Mia

D. Anna, Lea, Mia, Emma ***E.*** Mia, Anna, Emma, Lea

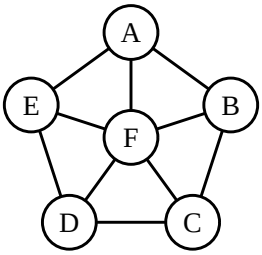
Part B: Each question is worth 4 points

4. Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned. Which of the pieces lies on the dot?



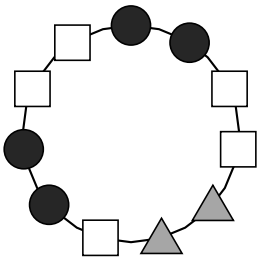
A. ***B.*** ***C.*** ***D.*** ***E.***

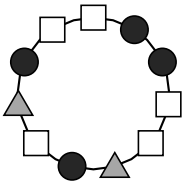
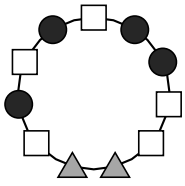
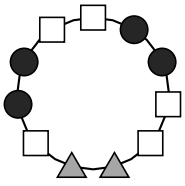
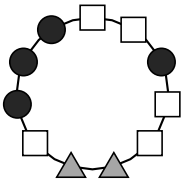
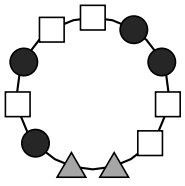
5. A zoo has 6 animals, A to F. A line between two animals means that they do not get along, so they must not live in the same cage. Any number of animals can live in one cage. What is the smallest number of cages the zoo needs?



A. 1 cage ***B.*** 2 cages ***C.*** 3 cages ***D.*** 4 cages ***E.*** 6 cages

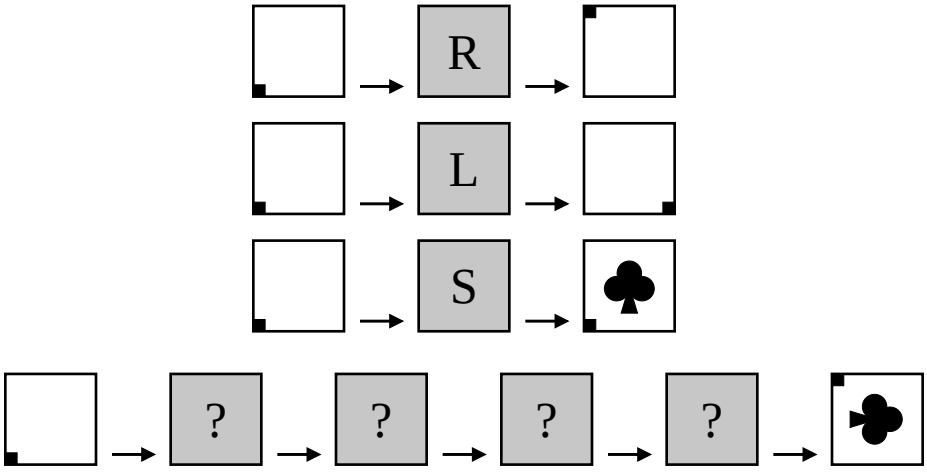
6. Anna has this bracelet with circles, squares and triangles and turns it around a bit. Which picture can show the bracelet now?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

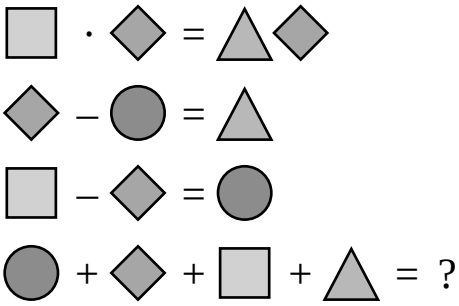
Part C: Each question is worth 5 points

7. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Machine L turns the sheet a quarter turn anticlockwise. Anna uses the machines 4 times in a row and gets the picture at the end of the bottom row. Which order works?



A. SRSS ***B.*** RRSL ***C.*** SSRL ***D.*** SLRR ***E.*** SSLR

8. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct. What number does Ben get for the last calculation?

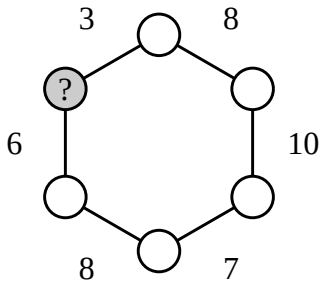


A. 13 ***B.*** 14 ***C.*** 15 ***D.*** 16 ***E.*** 17

9. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Zoe takes all the **heart-shaped** cookies that are still on the tray.
Tina takes all the **white** cookies that are still on the tray.
Mia takes all the **large** cookies that are still on the tray.
One of the girls takes 3 cookies, one takes 4 and one takes 9. Which of these groups of cookies did one of the girls take?

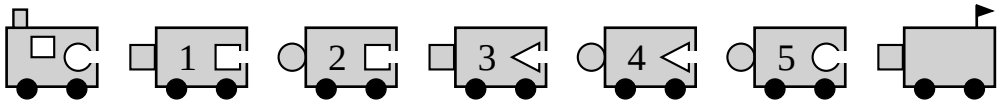
A. ***B.*** ***C.*** ***D.*** ***E.***

10. The numbers from 1 to 6 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



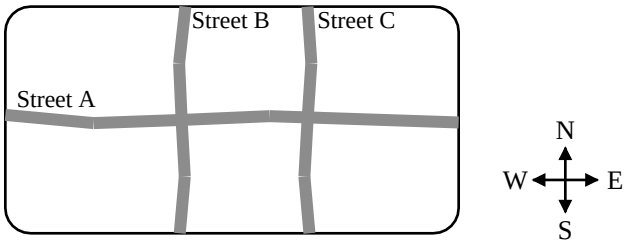
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

11. Anna has these pieces: an engine, a last wagon and 5 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. Anna has already put wagon 5 right after the engine. In how many different ways can Anna finish the train?



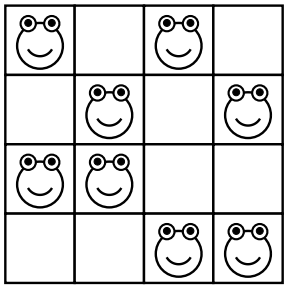
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

12. Streets B and C both go from north to south (see map). North of street A there are 12 houses. South of street A there are 18 houses. West of street B there are 11 houses. East of street C there are 8 houses. How many houses are there between street B and street C?



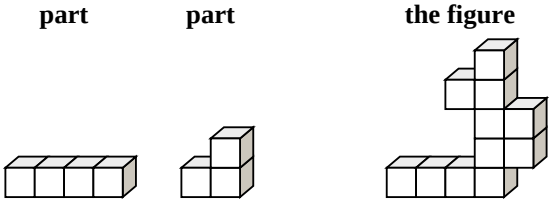
A. 6 houses ***B.*** 11 houses ***C.*** 13 houses ***D.*** 19 houses ***E.*** 22 houses

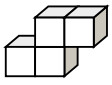
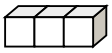
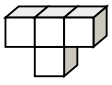
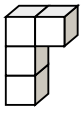
13. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two frogs jump at the same time, each to an empty neighbouring field, and after that there are again 2 frogs in each row and in each column. How many of the frogs can **never** be one of the two jumping frogs?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 8

14. Kanga built this flat figure out of 3 parts made of cubes, turned or flipped. Two of the parts are shown next to it. Which part is the third one?



A. ***B.***  ***C.***  ***D.***  ***E.*** 

15. A big lamp flashes every 4 seconds and a small lamp flashes every 10 seconds. They flash together right at the start. How many times do they flash **together** in the next 60 seconds? (Count a flash exactly at second 60, but not the one at the start.)

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 21

Answer Key

1.	**D**	**2.**	**B**	**3.**	**D**	**4.**	**B**	**5.**	**D**	**6.**	**C**
7.	**D**	**8.**	**E**	**9.**	**A**	**10.**	**A**	**11.**	**B**	**12.**	**B**
13.	**A**	**14.**	**D**	**15.**	**C**						

Solutions

1. D — Compare each pattern with all the others. Pattern D has exactly 4 grey cells in common with every other pattern, so it is the right one. [set-01/by_level/Grade_03-04_Ecolier/2023 Q24 · which_pattern]

2. B (5 times) — Between two meetings they run exactly once around the circle together. Tom only gets to the next letter (1 of 6 parts), so Mia runs the other 5 parts. 6 meetings × 5 parts = 30 parts = 5 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

3. D (Anna, Lea, Mia, Emma) — Check each order against every rule. Only Anna, Lea, Mia, Emma keeps all of them. [set-01/by_level/Grade_03-04_Ecolier/2023 Q19 · which_order]

4. B — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece B lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

5. D (4 cages) — A cage is like a colour: animals joined by a line need different cages. Look for animals that are all joined to each other. Then share the animals out step by step: 4 cages are enough, and fewer do not work. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · zoo_cages]

6. C — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only C has all 11 stones in the same order; in the others two stones have swapped places. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

7. D (SLRR) — Check each order: follow the corner mark through every machine and turn the club only with the machines after S. Only SLRR gives the picture. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · left_machine]

8. E (17) — Start with the calculation where the fewest digits are possible, then check all three: diamond = 5, square = 7, triangle = 3, circle = 2. 2 + 5 + 7 + 3 = 17. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]

9. A — Try the orders. Only Mia, Zoe and then Tina gives 3, 4 and 9: Mia takes 9, Zoe takes 4, Tina takes 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_group]

10. A (1) — With 6 circles every guess closes the ring, so check something else: the numbers must be 1 to 6, each once. Only 1 gives 1, 2, 6, 4, 3, 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · even_ring]

11. B (2) — Look only at the branch that starts with wagon 5: H–5–2–T, H–5–2–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

12. B (11 houses) — The village has 12 + 18 = 30 houses. Take away the houses west of B and east of C: 30 – 11 – 8 = 11 houses are left between the two streets. [set-01/by_level/Grade_03-04_Ecolier/2023 Q12 · three_streets]

13. A (0) — Find all the ways (15): the second jump must undo the first one (sideways between the same two columns, or up/down between the same two rows). 8 frogs take part in some way, so 8 - 8 = 0 never jump. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · never_jumps]

14. D — The figure has 11 cubes and the two parts have 7, so the third part has 4 cubes. Put the two parts into the figure; what is left has the shape of part D. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · missing_part]

15. C (3) — The big lamp flashes at 4, 8, 12, ... and the small one at 10, 20, 30, ... The first time both flash is at second 20 (not 4 × 10 = 40), then every 20 seconds. 60 ÷ 20 = 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · lamps]